DRESS: A Continuous Framework for Structural Graph Refinement

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Abstract

We introduce DRESS, a deterministic, parameter-free framework that iteratively refines the structural similarity of edges in a graph to produce a *canonical fingerprint*: a real-valued edge vector, obtained by converging a non-linear dynamical system to its unique fixed point. The fingerprint is *isomorphism-invariant* by construction, *numerically stable* (strictly bounded, precision-preserving, and mathematically well-posed), *fast* and *embarrassingly parallel* to compute: DRESS total runtime is $\mathcal{O}(I \cdot m \cdot d_{\max})$ for I iterations to convergence, and convergence is guaranteed by Birkhoff contraction. We generalize the original equation to Motif-DRESS (arbitrary structural motifs) and Generalized-DRESS (abstract aggregation template), and introduce Δ-DRESS, which runs DRESS on each vertex-deleted subgraph to boost expressiveness. Δ-DRESS empirically separates all 7,983 graphs in a comprehensive Strongly Regular Graph benchmark, and on the tested CFI instances ($k = 0, 1, 2, 3$), k-deletion (Δ^k-DRESS) empirically matches the (k+2)-WL boundary.

1 Introduction

A fundamental task in graph analysis is to assign to each graph a compact descriptor that is *canonical* (isomorphic graphs receive the same descriptor), *discriminative* (non-isomorphic graphs receive different descriptors whenever possible), and *cheap to compute*. Existing approaches fall into two broad categories: discrete combinatorial methods, such as the Weisfeiler–Leman (WL) hierarchy [10, 2], for which a naive refinement round at level k costs $\mathcal{O}(n^{k+1})$ and the total runtime is $\mathcal{O}(T_{k\text{-WL}} \cdot n^{k+1})$ after $T_{k\text{-WL}}$ rounds to stabilization; and learned representations, such as message-passing neural networks [11, 8], which require labeled training data and offer no worst-case guarantees.

We use the WL hierarchy as a standard expressiveness scale for refinement-based graph invariants; our comparison is relative to that scale and does not presuppose a complete characterization of the graphs distinguished by either WL or DRESS.

DRESS is a parameter-free, continuous dynamical system on *edges* that converges to a unique fixed point of real-valued structural similarities, producing a canonical fingerprint vector for any graph. The Original-DRESS equation [3] achieves this deterministically at $\mathcal{O}(m \cdot d_{\max})$ per iteration and $\mathcal{O}(I \cdot m \cdot d_{\max})$ total runtime after convergence, with numerical stability (all values in $[0, 2]$) and no learnable parameters. The expressiveness comparison with WL is discussed in Section 4 and the runtime comparison in Section 7.4.

We present a hierarchy of DRESS variants, building from the concrete to the general:

1. **Original-DRESS** (Section 2): The foundational equation, a parameter-free dynamical system on edges that converges to a canonical fingerprint vector.

2. **Expressiveness beyond 1-WL** (Section 3): Original-DRESS distinguishes graphs that 1-WL cannot, including the prism graph from $K_{3,3}$.

3. **Empirical Equivalence to 2-WL** (Section 4): Original-DRESS acts as a continuous empirical equivalent to the 2-dimensional Weisfeiler–Leman test.

4. **Motif-DRESS** (Section 5): A generalization replacing triangle neighborhoods with arbitrary structural motifs; Original-DRESS is the $M = K_3$ special case.

5. **Generalized-DRESS** (Section 6): The most abstract template, opening the aggregation function and norm as additional free parameters.

6. **Δ-DRESS** (Section 7): Runs DRESS on each vertex-deleted subgraph $G \setminus \{v\}$, connecting the framework to the reconstruction conjecture. Δ-DRESS empirically separates all 7,983 Strongly Regular Graphs in a comprehensive benchmark, and on the tested CFI instances ($k = 0, 1, 2, 3$), k-deletion (Δ^k-DRESS) empirically matches the (k+2)-WL boundary.

Isomorphism Test. Throughout this paper, the **graph fingerprint** is the sorted vector of converged edge values $\mathrm{sort}(d^*)$. Two graphs are declared non-isomorphic if and only if their fingerprints differ element-wise beyond a numerical tolerance ϵ. This applies uniformly to all DRESS variants. An equivalent representation is the **histogram** $h(d^*)$: in the unweighted case all DRESS values lie in $[0, 2]$ (see the convergence proof in Section 2.1) and the convergence tolerance is $\epsilon = 10^{-6}$, so each value maps to one of $\lceil 2/\epsilon \rceil = 2 \times 10^6$ integer bins, yielding a fixed-size bin-count vector that uniquely identifies the same multiset. In general, the number of bins is determined by the convergence tolerance: a tolerance of ϵ over the range $[0, 2]$ gives $\lceil 2/\epsilon \rceil$ bins. For weighted graphs, where values may exceed 2, the bin range is extended accordingly. The histogram is particularly useful when fingerprints from multiple subgraphs must be pooled (see Δ-DRESS, Section 7).

All experiments use a convergence tolerance of $\epsilon = 10^{-6}$; in every case tested, convergence was reached in fewer than 31 iterations.

Related Work. Higher-order GNN architectures that exceed 1-WL expressiveness include k-GNN [8] and PPGN [7]. Subgraph-based approaches such as GNN-AK+ [12] and ESAN [1] achieve higher expressiveness by running message-passing networks on vertex-deleted or vertex-marked subgraphs; Δ-DRESS (Section 7) shares the vertex-deletion strategy but replaces learned message passing with deterministic fixed-point iteration. All of the above are **supervised** methods that require labeled training data and learnable parameters. In contrast, DRESS is a deterministic, unsupervised framework that produces canonical fingerprints via fixed-point iteration, without any learning or parameter tuning.

2 Original-DRESS

The Original-DRESS equation [3] is a parameter-free, non-linear dynamical system that iteratively refines edge values based on common-neighbor aggregation.

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} \left(d_{ux}^{(t)} + d_{xv}^{(t)} \right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{1}$$

where $N[u] = N(u) \cup \{u\}$ denotes the closed neighborhood of u, and $\|u\| = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$ is the vertex norm. The self-similarity $d_{uu} = 2$ is invariant under iteration. This equation converges to a unique fixed point, providing a continuous structural fingerprint for the graph.

Self-loops. Self-loops are added to every vertex before iteration (i.e., the algorithm uses the closed neighborhood $N[u] = N(u) \cup \{u\}$). The self-loop edge (u, u) participates in both the aggregation and the vertex norm; without it, an isolated edge with no common neighbors would produce $\|u\| \cdot \|v\| = 0$, making the iteration undefined.

Initialization. The equation is scale-invariant (degree-0 homogeneous), so the fixed point is independent of the initial scale. Any uniform initialization $d_{uv}^{(0)} = c$ for all edges, with $c > 0$, converges to the same unique fixed point d^*. In practice, $c = 1$ is used.

However, because Original-DRESS aggregates strictly over triangles (common neighbors), it cannot distinguish graphs with identical triangle counts per edge (e.g., Strongly Regular Graphs).

2.1 Convergence

Theorem 1 (Convergence of Original-DRESS)**.** *Let $F\colon \mathbb{R}_+^{|E|} \to \mathbb{R}_+^{|E|}$ be the Original-DRESS update operator. Then F converges to a unique fixed point $d^* \in [0,2]^{|E|}$ for any uniform initial state $\mathbf{d}^{(0)} > 0$ (i.e., $\mathbf{d}^{(0)} = c\mathbf{1}$ for $c > 0$).*

Proof. The argument proceeds in three steps:

1. **Scale Invariance (Degree-0 Homogeneity):** The numerator $\sum_{x \in N[u] \cap N[v]} (d_{ux} + d_{xv})$ is positively homogeneous of degree $p = 1$ in $\mathbf{d}$, and each vertex norm $\|u\| = \sqrt{\sum_{x \in N[u]} d_{ux}}$ is homogeneous of degree $q = \frac{1}{2}$. The denominator $\|u\| \cdot \|v\|$ is therefore degree $2q = 1 = p$, making F homogeneous of degree 0: for any $\lambda > 0$, $F(\lambda \mathbf{d}) = F(\mathbf{d})$. The iteration is self-regularizing and cannot diverge or collapse to zero.

2. **Boundedness:** For any uniform positive initialization $\mathbf{d}^{(0)} = c\mathbf{1}$ ($c > 0$), the scale invariance of F implies that the first iterate reduces to the unit case: $F(c\mathbf{1}) = F(\mathbf{1})$. Specifically, $F(\mathbf{1})_{uv} = \frac{2|N[u] \cap N[v]|}{\sqrt{\deg(u)\deg(v)}}$. Since $|N[u] \cap N[v]| \leq \sqrt{\deg(u)\deg(v)}$ (Cauchy-Schwarz), $F(c\mathbf{1})_{uv} \leq 2$. The scale invariance and the topological structure of the operator ensure $[0,2]^{|E|}$ is a forward-invariant set for this orbit, meaning all subsequent iterates remain bounded in this interval.

3. **Contraction on the Hilbert Projective Metric:** F is a positive, degree-0 homogeneous map on the cone $\mathbb{R}_{>0}^{|E|}$. By Birkhoff's contraction theorem, F is a strict contraction under the Hilbert projective metric $d_H(x,y) = \log\!\left(\max_e \frac{x_e}{y_e} \cdot \max_e \frac{y_e}{x_e}\right)$, provided F maps a bounded part of the cone into a strictly smaller part, which follows from the boundedness above. By the Banach fixed-point theorem on $(\mathbb{R}_{>0}^{|E|}/\!\sim,\, d_H)$, the iteration converges to a unique ray, and the forward-invariant boundedness from the $\mathbf{d}^{(0)} = \mathbf{1}$ initialization pins the limit to a finite vector $d^* \in [0,2]^{|E|}$.

A complete formal verification of the contraction constant is deferred to future work; all empirical tests confirm convergence within 20 iterations. □

3 Original-DRESS Distinguishes Graphs That 1-WL Cannot

We now demonstrate a key expressiveness result: Original-DRESS, the simplest member of the DRESS family, already distinguishes graphs that 1-WL (color refinement) cannot.

Theorem 2 (DRESS distinguishes beyond 1-WL)**.** *There exist graph pairs that 1-WL cannot distinguish but Original-DRESS can. In particular, DRESS distinguishes the prism graph ($C_3 \square K_2$) from the complete bipartite graph $K_{3,3}$, a pair that 1-WL provably cannot separate.*

Proof. Both graphs are 3-regular on 6 vertices with 9 edges. Since all vertices have the same degree, 1-WL assigns a uniform color to every vertex in both graphs after every iteration, and therefore cannot distinguish them [2].

DRESS operates on *edges* and aggregates over common neighbors (triangles). We show that the prism must have at least two distinct edge values at convergence, while $K_{3,3}$ has a single uniform value.

$K_{3,3}$**:** No two adjacent vertices share a common neighbor ($K_{3,3}$ is triangle-free). Hence every edge has the same neighborhood structure, and by symmetry the unique fixed point assigns the same value to all 9 edges. Numerically, $d^*_{K_{3,3}} \approx 1.155$ for all edges.

Prism graph: There are two structurally distinct edge types: *triangle edges* (6 edges forming the two triangular faces; each pair of endpoints shares 1 common neighbor) and *matching edges* (3 edges connecting corresponding vertices of the two triangles; each pair of endpoints shares 0 common neighbors besides self-loops). Suppose for contradiction that all 9 edges converge to the same value d^*. Then:

- Triangle edges: $d^* = \dfrac{4+4d^*}{2+3d^*}$, which gives $d^* = \frac{1+\sqrt{13}}{3} \approx 1.535$.
- Matching edges: $d^* = \dfrac{4+2d^*}{2+3d^*}$, which gives $d^* = \frac{2}{\sqrt{3}} \approx 1.155$.

These are contradictory, so the prism cannot have a uniform fixed point. The converged unique edge values are:

Graph	**Unique values**	**Multiplicities**
Prism	$\{0.922,\ 1.709\}$	3 matching edges, 6 triangle edges
$K_{3,3}$	$\{2/\sqrt{3} \approx 1.155\}$	all 9 edges identical

Since 1-WL cannot distinguish them and DRESS can, DRESS distinguishes beyond 1-WL on this instance. □

Remark. This result holds for Original-DRESS alone, the simplest member of the DRESS family, using only triangle neighborhoods and $\mathcal{O}(m \cdot d_{\max})$ computation per iteration. The generalizations that follow (Motif-DRESS, Δ-DRESS) extend this advantage further to graphs that resist even 2-WL.

4 Original-DRESS as the Continuous Analogue of 2-WL

The previous section showed that DRESS distinguishes *specific* graph pairs that 1-WL cannot. We now establish the theoretical bounds of Original-DRESS: it acts as a continuous, differentiable analogue to the 2-dimensional Weisfeiler–Leman (2-WL) test.

Theorem 3 (Empirical Equivalence to 2-WL)**.** *In practical floating-point computation, Original-DRESS operates as an empirical continuous equivalent to* 2*-WL (Original-DRESS* $\equiv$ *2-WL).*

Proof. Both 2-WL and Original-DRESS update edge states by aggregating 3-node interactions (triangles). While 2-WL applies an injective hash to the discrete *multiset* of neighbor colors to branch lossless partitions, Original-DRESS aggregates these same neighborhoods via continuous summation: $\sum_x (d_{ux} + d_{xv})$. Because summation is technically a lossy operator over multisets, Original-DRESS is theoretically bounded by 2-WL (Original-DRESS $\leq$ 2-WL).

However, DRESS values are irrational limits of a nonlinear dynamical system. For a "sum collision" to occur between non-isomorphic structures, distinctly generated sets of irrationals must coincidentally sum to the exact same value. Over the reals $\mathbb{R}$, the probability of such an algebraic collision is exactly zero ($\mathbb{P} = 0$). Thus, up to numerical precision, the sums strictly preserve multiset distinctions, making Original-DRESS an exact empirical equivalent to 2-WL without the massive $\mathcal{O}(n^3)$ memory overhead of discrete hashing. □

Remark 4. *Section 9.3 shows empirically that* Δ^k*-DRESS* $\geq (k+2)$*-WL: each deletion level adds exactly one WL dimension of expressiveness.*

5 Motif-DRESS

Original-DRESS is limited to triangle neighborhoods. What if we use other structural motifs? Motif-DRESS generalizes Original-DRESS by replacing the common-neighbor aggregation with a motif-defined symmetric vertex neighborhood $\mathcal{N}_M(u,v) = \mathcal{N}_M(v,u)$: the set of endpoints of edges in instances of a motif M containing edge (u,v) that are adjacent to u or v, always including u and v themselves, with an optional symmetric weight function $\bar{w} : E \to \mathbb{R}_{>0}$ (with $\bar{w}_e = 1$ for unweighted graphs).

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in \mathcal{N}_M(u,v)} \left(\bar{w}_{ux} \cdot d_{ux}^{(t)} + \bar{w}_{xv} \cdot d_{xv}^{(t)}\right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{2}$$

where $\|u\|^{(t)} = \sqrt{\sum_{x \in N[u]} \bar{w}_{ux} \cdot d_{ux}^{(t)}}$, and $\bar{w}_{ab} \cdot d_{ab} = 0$ whenever $(a,b) \notin E$. The symmetric weight function $\bar{w}(e)$ acts as a multiplicative factor, controlling how much structural information flows along each edge. Because the weights appear identically in the numerator and denominator (both are degree-1 in $\bar{w} \cdot d$), uniformly scaling all weights does not change the fixed point; only the relative weights matter.

Different choices of motif M yield different neighborhoods:

- **Triangle ($M = K_3$):** $\mathcal{N}_{K_3}(u,v) = N[u] \cap N[v]$, the closed common neighborhood. This recovers Original-DRESS exactly: $\sum_{x \in N[u] \cap N[v]} \left(\bar{w}_{ux}\, d_{ux} + \bar{w}_{xv}\, d_{xv}\right)$.
- **K_4 clique:** For each pair x, y with $\{u,v,x,y\}$ forming a K_4, $\mathcal{N}_{K_4}(u,v)$ contains u, v, x, and y. Each vertex contributes $\bar{w}_{ux}\, d_{ux} + \bar{w}_{xv}\, d_{xv}$.
- **4-cycle ($M = C_4$):** For each 4-cycle u-x-y-v, $\mathcal{N}_{C_4}(u,v)$ contains u, v, x, and y. Vertex x contributes $\bar{w}_{ux}\, d_{ux}$; vertex y contributes $\bar{w}_{yv}\, d_{yv}$ (cross-terms are zero since $(x,v), (u,y) \notin E$ in the C_4).

5.1 Properties

Complexity: $\mathcal{O}(\text{Motif Extraction}) + \mathcal{O}\left(I \cdot \sum_e |\mathcal{N}_M(e)|\right)$, where I is the number of iterations and $\sum_e |\mathcal{N}_M(e)|$ is the total size of all motif neighborhoods. For Original-DRESS (triangles), this reduces to $\mathcal{O}(I \cdot m \cdot d_{\max})$. For motifs such as 4-cycles or K_4 cliques in sparse graphs, the motif neighborhoods can be significantly smaller, making Motif-DRESS faster than Original-DRESS.

Invariant: $d_{uu} = 2$. The self-similarity $d_{uu} = 2$ is a constant maintained throughout iteration. Since the iteration only updates d_{uv} for $u \neq v$, and the norm $\|u\| = \sqrt{\sum_{x \in N[u]} \bar{w}_{ux} \cdot d_{ux}}$ always includes $d_{uu} = 2$, this is a fixed property of the equation, not a free parameter.

5.2 Expressiveness

Bypassing WL Limitations: Standard message-passing architectures and 2-WL update edge states based exclusively on the intersection of localized vertex neighborhoods (effectively tracking K_3 structures). However, higher-order structures like the Rook and Shrikhande graphs famously resist even the 3-WL test.

By precomputing higher-order structural motifs like K_4 and using them to define the aggregation neighborhood, Motif-DRESS bypasses these WL barriers. It explicitly injects these topological invariants into the continuous message-passing framework. Because the motif extraction acts as a one-time preprocessing step, Motif-DRESS achieves strictly greater expressiveness than 3-WL on these theoretical graphs while retaining identical iterative computational complexity ($\mathcal{O}(I \cdot \sum_e |\mathcal{N}_M(e)|)$).

All experiments below use the K_4 clique motif. The specific SRG pairs tested below are known to be indistinguishable by either 2-WL or 3-WL; each successful distinction therefore demonstrates that Motif-DRESS empirically exceeds these corresponding WL boundaries.

- **Rook vs. Shrikhande:** Successfully distinguishes this pair of SRGs with parameters $(16, 6, 2, 2)$, which are indistinguishable by 3-WL. The Rook graph ($K_4 \square K_4$) contains K_4 cliques while the Shrikhande graph does not, so the K_4-neighborhood sizes differ per edge.
- **Chang Graphs:** Distinguishes 3 of the 6 pairwise comparisons among the four SRGs with parameters $(28, 12, 6, 4)$: T(8) vs each of Chang-1, Chang-2, and Chang-3. The three Chang graphs are pairwise indistinguishable by Motif-K_4 (all three have identical K_4-neighborhood structure per edge).

5.3 Convergence

The convergence proof for Original-DRESS (Theorem 1) generalizes directly to Motif-DRESS. The same three sufficient mechanisms apply: degree-0 homogeneity ($p = 2q$), initial normalization into a strictly bounded forward-invariant set from a uniform positive state $\mathbf{d}^{(0)} = c\mathbf{1}$, and Birkhoff contraction on the Hilbert projective metric. These hold for any motif neighborhood $\mathcal{N}_M$ and symmetric non-negative weight function $\bar{w}$. For unweighted motifs, values are bounded in $[0, 2]$; with arbitrary edge weights, the finite initial bounds scale proportionally but strictly remain mathematically bounded, ensuring guaranteed convergence to potentially higher fixed point values.

6 Generalized-DRESS

Motif-DRESS fixes the aggregation to summation and the norm to the product of geometric means. Generalized-DRESS is the most abstract template, allowing any choice of these components as long as the resulting update rule preserves the convergence guarantees (degree-0 homogeneity, boundedness, and contraction; see Section 2.1). For each edge (u, v):

$$d^{(t+1)} = \frac{f\big(\mathbf{d}^{(t)}, \mathcal{N}, \bar{w}\big)}{g\big(\mathbf{d}^{(t)}, \mathcal{N}, \bar{w}\big)} \tag{3}$$

where $d \equiv d_{uv}$ is the similarity value assigned to edge (u, v), and:

- $\mathcal{N}(u, v)$ is a **symmetric neighborhood operator**, the structural context aggregated for (u, v),
- $\bar{w} : E \to \mathbb{R}_{>0}$ is a **symmetric weight function** ($\bar{w}(u, v) = \bar{w}(v, u)$; $\bar{w} \equiv 1$ for unweighted graphs),
- f is the **aggregation function**,
- g is the **norm function**.

Because $\mathcal{N}$ and $\bar{w}$ are symmetric, f and g receive the same inputs for (u, v) and (v, u), so $d(u, v) = d(v, u)$ holds for every member of the family. For Original-DRESS and Motif-DRESS this follows directly from the equation; in the general case it is guaranteed by the symmetry of the inputs.

Original-DRESS and Motif-DRESS are both special cases: Original-DRESS fixes $\mathcal{N}$ to triangles, $f = \text{sum}$, $g = \|u\| \cdot \|v\|$ (product of geometric means), $\bar{w} \equiv 1$; Motif-DRESS generalizes $\mathcal{N}$ to arbitrary motifs and $\bar{w}$ to non-uniform weights while keeping the same f and g. Generalized-DRESS opens all four parameters, enabling variants such as Cosine-DRESS (cosine similarity aggregation) or Minkowski-r norms.

7 Δ-DRESS

Δ-DRESS breaks symmetry by running DRESS on each vertex-deleted subgraph $G \setminus \{v\}$ for every $v \in V$. The Δ-DRESS fingerprint is the multiset of per-vertex DRESS fingerprints, or equivalently a pooled histogram accumulating all converged edge values across all n deletions.

Deleting a vertex from a regular graph produces an irregular subgraph where DRESS can now distinguish structure that was hidden by the uniform regularity.

7.1 Connection to the Reconstruction Conjecture

The multiset $\{\{\text{DRESS}(G \setminus \{v\}) : v \in V\}\}$ is directly analogous to the *deck* in the Kelly–Ulam reconstruction conjecture [6, 9], which posits that graphs with $n \geq 3$ are determined (up to isomorphism) by their multiset of vertex-deleted subgraphs. Δ-DRESS computes a continuous relaxation of this deck.

7.2 Expressiveness

Δ-DRESS empirically distinguishes the following non-isomorphic pairs:

- **Rook vs. Shrikhande:** Successfully distinguished ($\text{SRG}(16, 6, 2, 2)$; confounds 2-WL).
- **Chang Graphs:** Distinguished all 6 pairs among T(8) and the three Chang graphs ($\text{SRG}(28, 12, 6, 4)$; confound 2-WL).
- $2 \times C_4$ **vs.** C_8**:** Successfully distinguished (both 2-regular on 8 vertices).
- **Petersen vs. Pentagonal Prism:** Successfully distinguished (both 3-regular on 10 vertices).

7.3 Δ^k-DRESS

We now define the k-deletion operator formally.

Definition 5 (Δ^k-DRESS)**.** *For a graph $G = (V, E)$, a DRESS variant $\mathcal{F}$, and a deletion depth $k \geq 0$, the Δ^k-DRESS fingerprint is the multiset of per-deletion sorted edge-value vectors:*

$$\Delta^k\text{-}DRESS(\mathcal{F}, G) = \{\{\text{sort}(\mathcal{F}(G \setminus S)) : S \subset V,\ |S| = k\}\}$$

where $G \setminus S$ is the subgraph induced by $V \setminus S$ and $\mathcal{F}(G \setminus S)$ is the converged DRESS edge-value vector.

Definition 6 (Histogram fingerprint)**.** *For a graph $G = (V, E)$, a DRESS variant $\mathcal{F}$, and a deletion depth $k \geq 0$, the histogram fingerprint of Δ^k-DRESS is:*

$$h\Big(\Delta^k\text{-}DRESS(\mathcal{F}, G)\Big) = h\left(\bigsqcup_{S \subset V,\ |S|=k} \mathcal{F}(G \setminus S)\right)$$

where h maps each edge value to an integer bin of width ϵ, producing a fixed-size bin-count vector of $\lceil 2/\epsilon \rceil$ bins.

Proposition 7 (Isomorphism Invariance)**.** *Δ^k-DRESS is an isomorphism invariant: if $G \cong H$ via an isomorphism $\phi : V(G) \to V(H)$, then*

$$\Delta^k\text{-}DRESS(\mathcal{F}, G) = \Delta^k\text{-}DRESS(\mathcal{F}, H).$$

Proof. Any isomorphism $\phi : G \to H$ induces a bijection on k-element vertex subsets: $S \mapsto \phi(S)$. For each $S \subset V(G)$ with $|S| = k$, the restriction $\phi|_{V(G)\setminus S}$ is an isomorphism $G \setminus S \xrightarrow{\sim} H \setminus \phi(S)$. Since DRESS depends only on graph structure (not vertex labels), we have $\mathcal{F}(G \setminus S) = \mathcal{F}(H \setminus \phi(S))$ as multisets of edge values. The bijection $S \leftrightarrow \phi(S)$ matches every term of the multiset union on one side to an equal term on the other:

$$\bigsqcup_{S \subset V(G),\, |S|=k} \mathcal{F}(G \setminus S) \;=\; \bigsqcup_{T \subset V(H),\, |T|=k} \mathcal{F}(H \setminus T).$$

The bijection $S \leftrightarrow \phi(S)$ therefore gives Δ^k-DRESS$(\mathcal{F}, G) = \Delta^k$-DRESS$(\mathcal{F}, H)$. Since h depends only on the multiset of edge values, the histogram fingerprint $h(\Delta^k\text{-DRESS}(\mathcal{F}, G)) = h(\Delta^k\text{-DRESS}(\mathcal{F}, H))$ is likewise an isomorphism invariant. □

7.4 Complexity

The time complexity of Δ^k-DRESS$(\mathcal{F}, G)$ is $\binom{n}{k}$ times the complexity of a single DRESS run:

$$\mathcal{O}\left(\binom{n}{k} \cdot T_{\mathcal{F}}\right)$$

where $T_{\mathcal{F}}$ is the runtime of the chosen DRESS variant $\mathcal{F}$ on a single subgraph. The $\binom{n}{k}$ runs are entirely independent, making Δ^k-DRESS embarrassingly parallel.

For Original-DRESS, $T_{\mathcal{F}} = \mathcal{O}(I \cdot m \cdot d_{\max})$, giving a total of $\mathcal{O}\big(\binom{n}{k} \cdot I \cdot m \cdot d_{\max}\big)$. For $k = 1$ this is $\mathcal{O}(n \cdot I \cdot m \cdot d_{\max})$; for $k = 0$ there is a single run on G itself, recovering Original-DRESS at $\mathcal{O}(I \cdot m \cdot d_{\max})$.

For comparison, one naive refinement round of $(k{+}2)$-WL costs $\mathcal{O}(n^{k+3})$; after $T_{(k+2)\text{-WL}}$ rounds to stabilization, the total runtime is $\mathcal{O}\big(T_{(k+2)\text{-WL}} \cdot n^{k+3}\big)$. The full multiset fingerprint of Δ^k-DRESS requires $\mathcal{O}\big(\binom{n}{k} \cdot m\big)$ space; the histogram fingerprint reduces this to $\mathcal{O}(n + m + \lceil 2/\epsilon \rceil)$ by processing one subgraph at a time and accumulating edge values into a fixed-size bin array, both compared to $\mathcal{O}(n^{k+2})$ for storing $(k{+}2)$-WL colors over all tuples.

8 Family Structure

The DRESS variants introduced in this paper form a nested hierarchy with one orthogonal composition operator:

$$\text{Generalized-DRESS} \supset \text{Motif-DRESS} \supset \text{Original-DRESS}$$

Δ^k is an **orthogonal wrapper** applicable to any of the above.

9 Experimental Results

All experiments use convergence tolerance $\epsilon = 10^{-6}$. For Δ-DRESS, two fingerprint representations are compared: the **multiset fingerprint** (the full sorted concatenation of all per-deletion DRESS vectors) and the **histogram fingerprint**. The multiset representation preserves all numerical information; the histogram is a fixed-size summary that is faster to compare and more convenient for pooling across deletions.

9.1 Convergence on Real-World Graphs

Table 1 reports convergence on real-world graphs spanning four orders of magnitude in size ($\epsilon = 10^{-6}$, max 100 iterations).

Graph	$\lvert V\rvert$	$\lvert E\rvert$	**Iter.**	**Final** δ
Wiki-Vote	8,298	103,689	17	8.31×10^{-7}
Amazon co-purchasing	548,552	925,872	18	6.35×10^{-7}
LiveJournal	4,033,138	27,933,062	30	7.09×10^{-7}
Facebook (KONECT)	59,216,215	92,522,012	26	6.84×10^{-7}

Table 1: DRESS convergence iterations and final residual $\delta = \max_e |d_e^{(t)} - d_e^{(t-1)}|$. Even on graphs with 59 M vertices, convergence requires fewer than 31 iterations. Across these benchmarks, convergence was reached in 17 to 30 iterations.

9.2 Strongly Regular Graphs

Strongly Regular Graphs (SRGs) are the canonical hard instances for polynomial-time isomorphism methods: all edges share identical local structure (same degree, same common-neighbor counts), so Original-DRESS (Δ^0) assigns the same value to every edge and produces a uniform fingerprint. This is expected: SRGs are regular, and the Original-DRESS iteration sees no local variation.

Δ^1-DRESS overcomes this limitation. By running DRESS on each vertex-deleted subgraph $G \setminus \{v\}$, the uniform regularity is broken and structurally distinct edges emerge. We tested Δ^1-DRESS on 7,983 strongly regular graphs from the repository of Krystal Guo [5], spanning three parameter families:

Family	**Parameters**	**Graphs**	**Separated**	**Min** L^∞
Conference (Mathon)	$(45, 22, 10, 11)$	6	**100%**	4.16×10^{-3}
Steiner S(2,4,28)	$(63, 32, 16, 16)$	4,466	**100%**	1.95×10^{-3}
Quasi-symmetric 2-designs	$(63, 32, 16, 16)$	3,511	**100%**	2.23×10^{-3}

All 7,983 graphs are pairwise distinguished by Δ^1-DRESS. The "Min L^∞" column reports the smallest element-wise maximum difference between the fingerprints of any sampled pair (1,000 random pairs per family). Values around 10^{-3} confirm that separations are genuine, not floating-point artifacts. This was further validated by checking that the unique count remains stable across all rounding precisions from 6 to 14 decimal digits.

Multiset vs. histogram fingerprint. Both representations produce identical separation results on all 7,983 SRGs. The multiset fingerprint preserves full numerical precision and is the canonical representation. The histogram fingerprint, with bin width $\epsilon = 10^{-6}$ over $[0, 2]$, maps each value to one of 2×10^6 integer bins. On these families the two representations agree perfectly.

9.3 CFI Staircase

The Cai–Fürer–Immerman (CFI) construction [2] produces the canonical hard instances for the WL hierarchy: distinguishing $\mathrm{CFI}(K_n)$ from $\mathrm{CFI}'(K_n)$ requires at least $(n-1)$-WL. We tested Δ^k-DRESS for $k = 0, 1, 2, 3$:

Base	$\lvert V\rvert$	**WL req.**	Δ^0	Δ^1	Δ^2	Δ^3
K_3	6	2-WL	✓	✓	✓	✓
K_4	16	3-WL	×	✓	✓	✓
K_5	40	4-WL	×	×	✓	✓
K_6	96	5-WL	×	×	×	✓
K_7	224	6-WL	×	×	×	×

For the tested depths $k = 0, 1, 2, 3$, the pattern is exact: each deletion level adds one WL dimension of expressiveness. Δ^k-DRESS distinguishes CFI(K_{k+3}) (requiring (k+2)-WL) and fails on CFI(K_{k+4}) (requiring (k+3)-WL), empirically matching the (k+2)-WL boundary on these instances.

9.4 DRESS–WL Dominance Conjecture

The experimental evidence above, together with the established practical equivalence to 2-WL (Theorem 3) and the CFI staircase results, motivates the following:

Remark 8 (DRESS–WL Continuous Dominance Conjecture)**.** *In practical continuous computation, Δ^k-DRESS acts as an empirical superset or equivalent to $(k+2)$-WL for all $k \geq 0$.*

For each fixed depth k, Δ^k-DRESS defines a precise graph invariant and hence a precise indistinguishability relation, just as k-WL does. The unresolved issue is the comparison theorem between these two equivalence notions.

Several structural properties of DRESS make this conjecture reasonable:

1. **Continuous vs. discrete.** Where WL produces discrete color classes, DRESS produces real-valued invariants. This yields a finer-grained representation of structural information and can preserve distinctions that discrete partitions collapse.

2. **Edges vs. vertices.** DRESS operates natively on edges, while 1-WL operates on vertices and 2-WL on vertex pairs. This places DRESS structurally closer to pair-based refinement methods than to vertex-only refinement.

3. **Non-linear update.** The DRESS update is a non-linear ratio rather than a discrete hash refinement, allowing it to encode interactions between neighborhood statistics in a different way from standard WL updates.

4. **Deletion as symmetry breaking.** Vertex deletion in Δ^k-DRESS plays a symmetry-breaking role analogous to individualization in higher-order WL arguments, providing a natural explanation for the observed expressiveness gains.

The base case ($k = 0$) is established as an empirical equivalence (Theorem 3). The properties of the inductive step ($k \geq 1$) remain an open conjecture.

10 Conclusion

We presented DRESS, a deterministic, parameter-free framework that assigns to any graph a canonical fingerprint vector in continuous space. The fingerprint is isomorphism-invariant, numerically stable ($d^* \in [0, 2]^{|E|}$), and computed at $\mathcal{O}(m \cdot d_{\max})$ per iteration with guaranteed convergence (Theorem 1). As a consequence of its continuous mathematical structure, Original-DRESS acts as an exact, memory-efficient empirical equivalent to 2-WL (Theorem 3), with per-iteration cost $\mathcal{O}(m \cdot d_{\max})$ versus the naive per-round cost $\mathcal{O}(n^3)$ of 2-WL. We generalized the original equation to Motif-DRESS (arbitrary structural motifs) and Generalized-DRESS (abstract aggregation template), and introduced Δ-DRESS, which breaks symmetry by vertex deletion. Δ-DRESS empirically separates all 7,983 Strongly Regular Graphs in a comprehensive benchmark, and on the tested CFI instances ($k = 0, 1, 2, 3$), k-deletion (Δ^k-DRESS) empirically matches the (k+2)-WL boundary. Because DRESS produces a canonical fingerprint that encodes structural similarity at every scale, it serves as a principled, scalable, parameter-free descriptor for downstream graph analysis tasks. This has already been demonstrated in community detection [3, 4], where DRESS edge values naturally classify intra- and inter-community edges. An open-source implementation with bindings for C, C++, Python, Rust, Go, Julia, R, MATLAB, and WebAssembly is publicly available at `https://github.com/velicast/dress-graph`.

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